Week 7: Student Performance – Data Cleaning & Transformation

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1. Introduction

The Student Performance dataset contains information about students, their demographic background, educational factors, and examination scores. The objective of this task is to clean, transform, and prepare the dataset using Power BI for further analysis and visualization.

2. Objective

- To import and inspect the Student Performance dataset.
- To clean and transform the dataset using Power Query.
- To create calculated columns such as Total Score, Average Score, and Performance Grade.
- To classify students based on Pass/Fail criteria.
- To reshape subject-wise score data using Unpivot.
- To create DAX measures for performance analysis.
- To prepare the dataset for visualization.

3. Dataset Description

The dataset contains student-related attributes such as:

- Student ID
- Gender
- Race/Ethnicity
- Parental Level of Education
- Lunch
- Test Preparation Course
- Math Score
- Reading Score

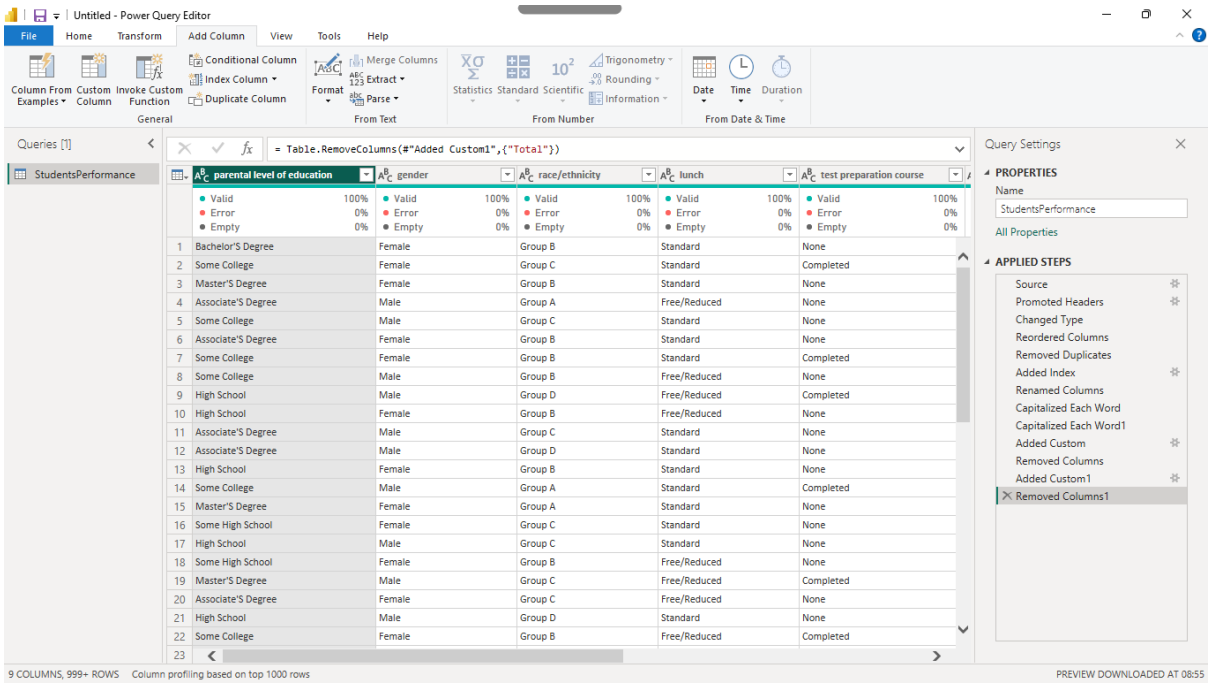
- Writing Score

These attributes were used to analyze students' academic performance and identify patterns in their examination results.

4. Data Import and Inspection

The Student Performance dataset was imported into Power BI using the **Get Data** option. The data was opened in **Power Query Editor** using Transform Data.

The columns, data types, missing values, errors, and duplicate records were inspected to understand the quality and structure of the dataset.



5. Data Cleaning

The dataset was cleaned and prepared for analysis by checking duplicate records, missing or invalid values, and data types. Numerical columns were converted into appropriate numeric data types, while categorical columns were reviewed for consistency.

The column names were also reviewed and standardized where required.

6. Categorical Data Transformation

The following categorical attributes were reviewed and standardized:

- Gender
- Race/Ethnicity

- Parental Level of Education
- Lunch
- Test Preparation Course

This helped maintain consistency in the dataset and made the categorical fields suitable for analysis.

7. Calculated Columns

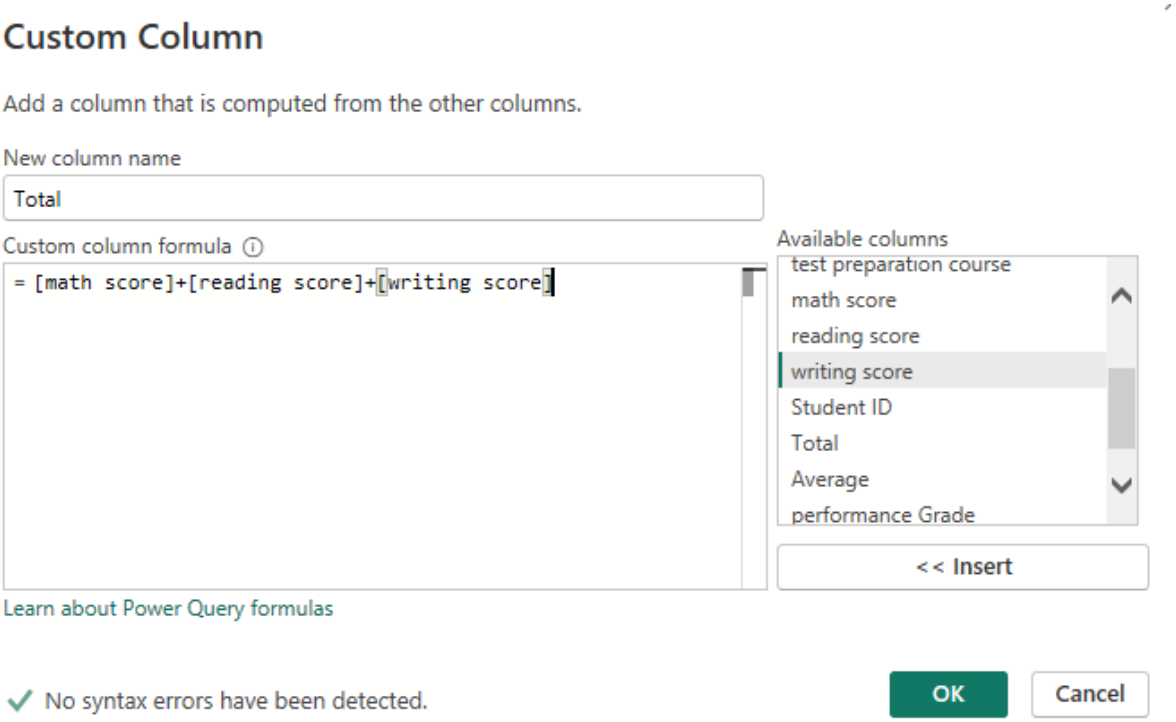
Three calculated columns were created using Power Query.

Total Score

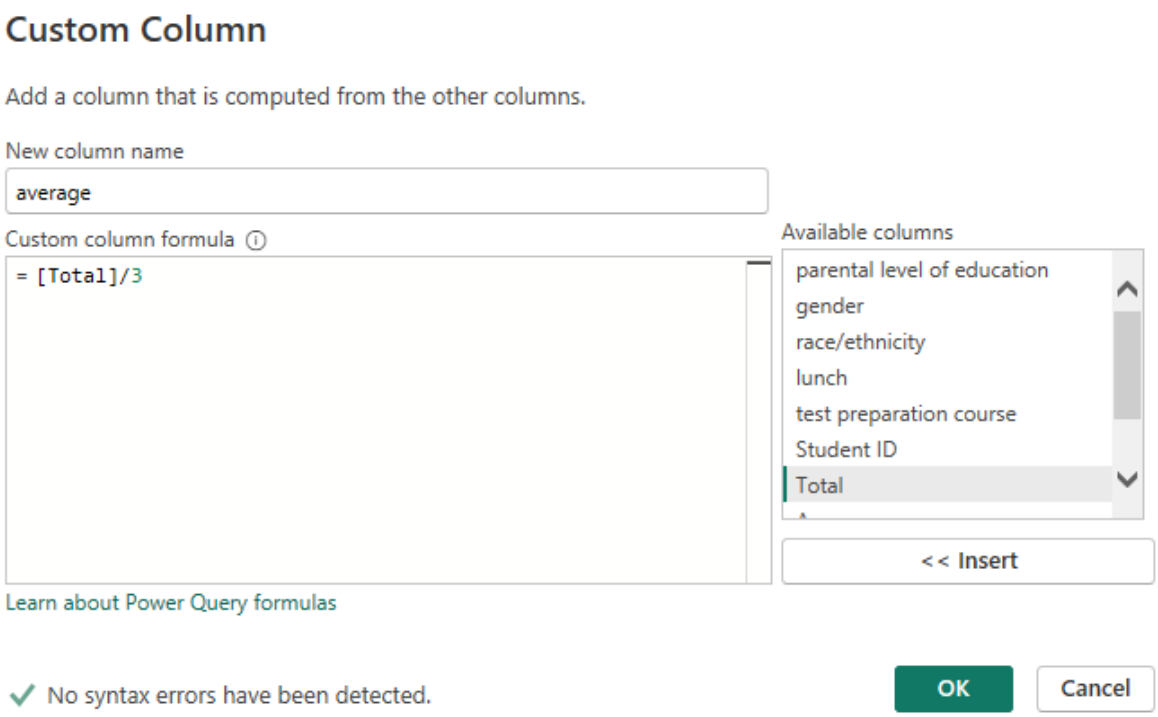
Total Score was calculated by adding the marks obtained in Math, Reading, and Writing.

Formula:

Total Score = Math Score + Reading Score + Writing Score



Average Score



Average Score was calculated by dividing the Total Score by 3.

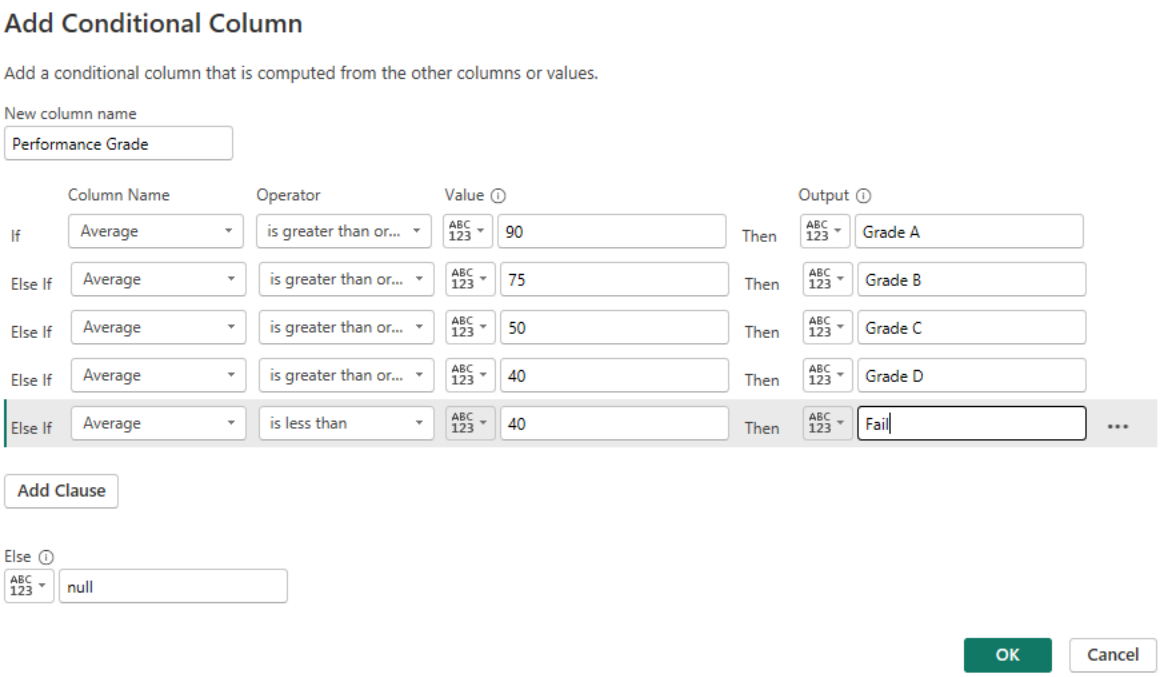
Formula:

Average Score = Total Score / 3

Performance Grade

A Performance Grade was created using conditional logic based on the Average Score.

- **A** – High Performance
- **B** – Good Performance
- **C** – Average Performance
- **F** – Low Performance

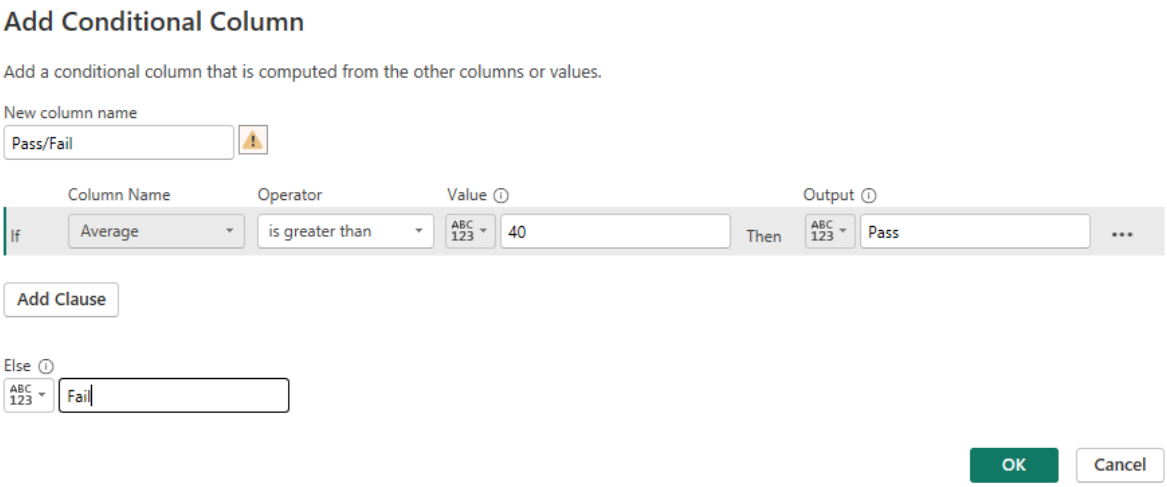


8. Pass/Fail Classification

A Pass/Fail classification was created using a suitable pass criterion based on the students' subject scores.

The students were classified into:

- **Pass**
- **Fail**



This classification was later used to calculate the overall pass rate using DAX.

9. Reshaping Subject Data

The Math Score, Reading Score, and Writing Score columns were selected and transformed using the **Unpivot Columns** option in Power Query.

The resulting columns were renamed as:

- **Subject**
- **Score**

Student ID	Attribute	Value
Valid 100%	Valid 100%	Valid 100%
Error 0%	Error 0%	Error 0%
Empty 0%	Empty 0%	Empty 0%
1	math score	72
1	reading score	72
1	writing score	74
2	math score	69
2	reading score	90
2	writing score	88
3	math score	90
3	reading score	95
3	writing score	93
4	math score	47
4	reading score	57
4	writing score	44
5	math score	76
5	reading score	78
5	writing score	75
6	math score	71
6	reading score	83
6	writing score	78
7	math score	88
7	reading score	95
7	writing score	92

Student ID	Subject	Score
Valid 100%	Valid 100%	Valid 100%
Error 0%	Error 0%	Error 0%
Empty 0%	Empty 0%	Empty 0%
1	math score	72
1	reading score	72
1	writing score	74
2	math score	69
2	reading score	90
2	writing score	88
3	math score	90
3	reading score	95
3	writing score	93
4	math score	47
4	reading score	57
4	writing score	44
5	math score	76
5	reading score	78
5	writing score	75
6	math score	71
6	reading score	83
6	writing score	78
7	math score	88
7	reading score	95
7	writing score	92

This transformation allows the performance of students to be analyzed subject-wise using a common Score column.

10. Data Model Preparation

After completing the transformations, the dataset was reviewed and unnecessary fields were removed where required. The final table was checked to ensure that the data structure was suitable for analysis.

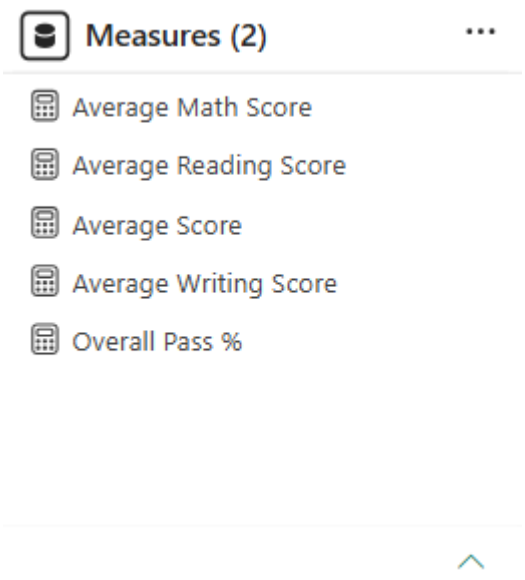
The transformed dataset was then loaded into Power BI using **Close & Apply**.

11. DAX Measures

The following DAX measures were created to evaluate overall student performance:

- Average Math Score
- Average Reading Score
- Average Writing Score

- Overall Average Score
- Overall Pass %



These measures provide summarized information about student academic performance.

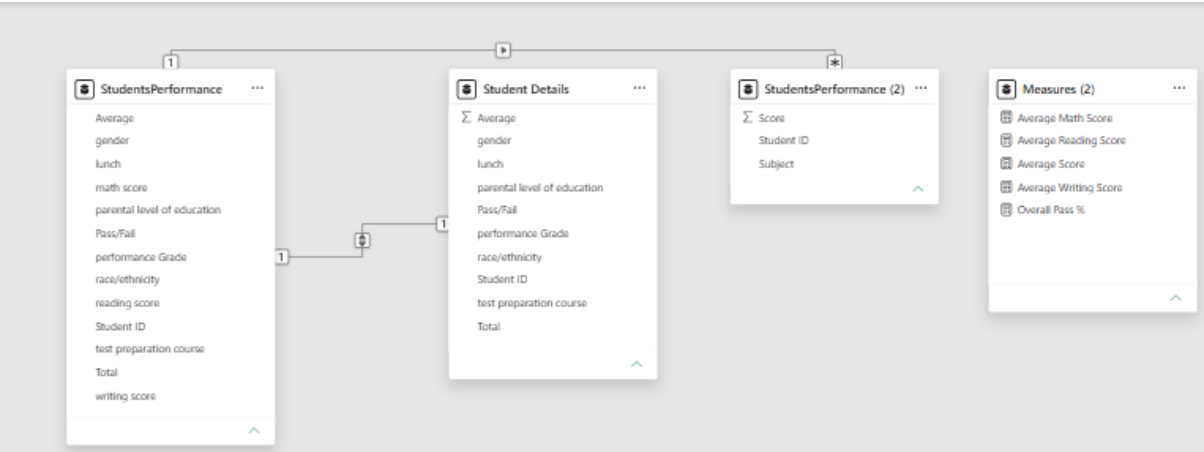
12. Validation

The transformed data was validated by comparing the calculated values with the original dataset. Total Score and Average Score calculations were checked, and the Pass/Fail and Performance Grade classifications were verified.

The unpivoted subject and score values were also checked to ensure that the transformation was performed correctly.

13. Result

The Student Performance dataset was successfully cleaned, transformed, and structured using Power BI. Calculated columns, Pass/Fail classification, Performance Grade, and DAX measures were created successfully. The subject-wise data was also reshaped using Unpivot, making the dataset ready for further visualization and analysis.



14. Conclusion

The Week 7 task successfully demonstrated the use of **Power Query for data cleaning and transformation** and **DAX for creating analytical measures**. The final dataset is structured and ready for creating meaningful visualizations and analyzing student performance.